Class 05: Data Viz wth ggplot

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Background

There are many graphics systems in R for making plots and figures. These include "base" graphics and "ggplot2". The first is older and add-on package **ggplot2**

Let's compare how we make a simple graph with these two systems:

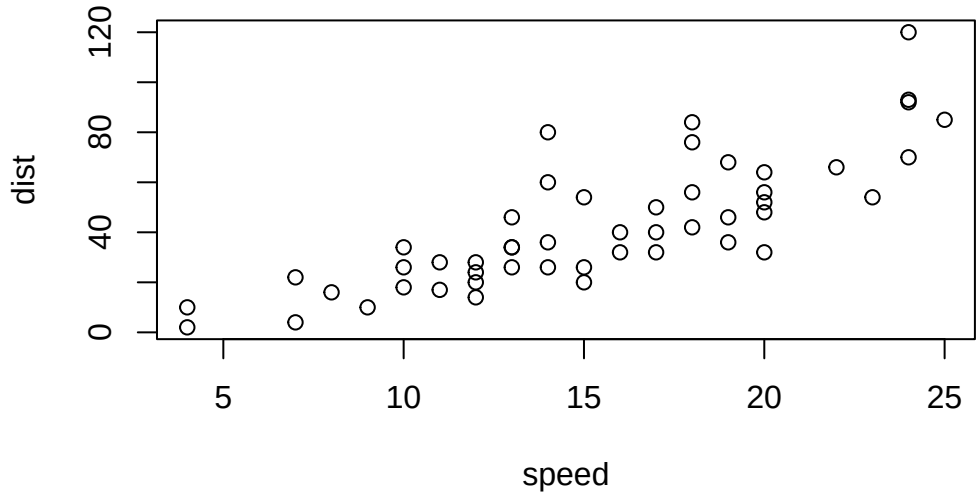
We can use the built-in `cars` dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

make a plot:

```
plot(cars)
```



Now for "ggplot2"

Before I can use ggplot2 I need to install it on my computer, to do this we can use function `install.packages("ggplot2")`

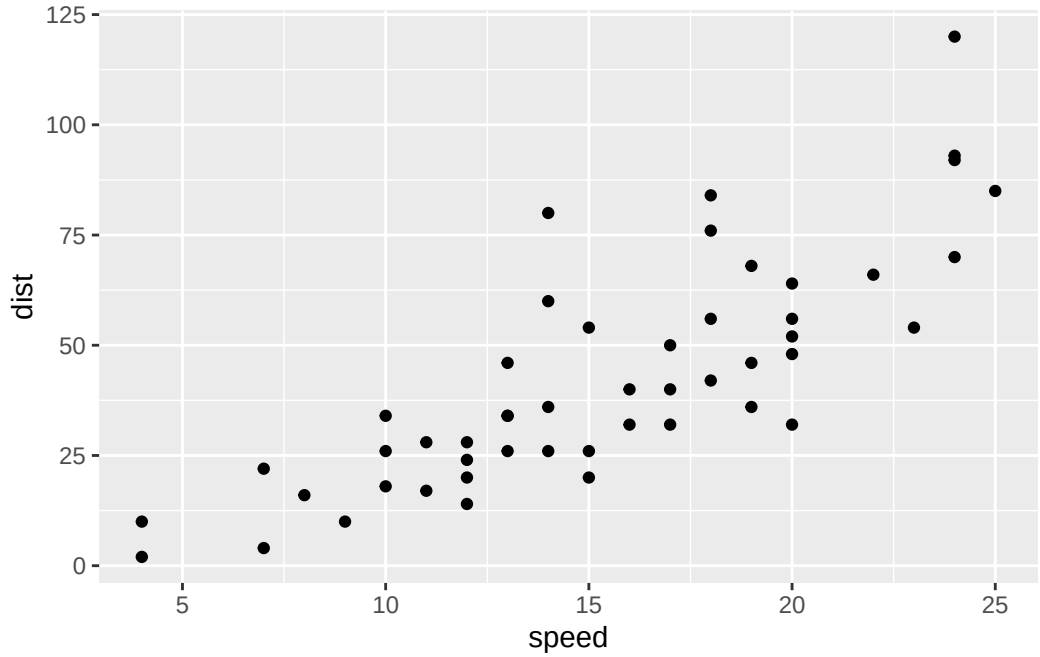
Once installed, we still need to load it into our R session. We do this with a "library()" call

```
library(ggplot2)
ggplot(cars)
```

Every ggplot needs at least 3 inputs

-the **data** (a data.frame of the stuff we want to plot), -the **aesthestics** (how the data maps to the plot), -the **geom** layer (how you want the plot drawn, prints, lines etc.)

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point()
```

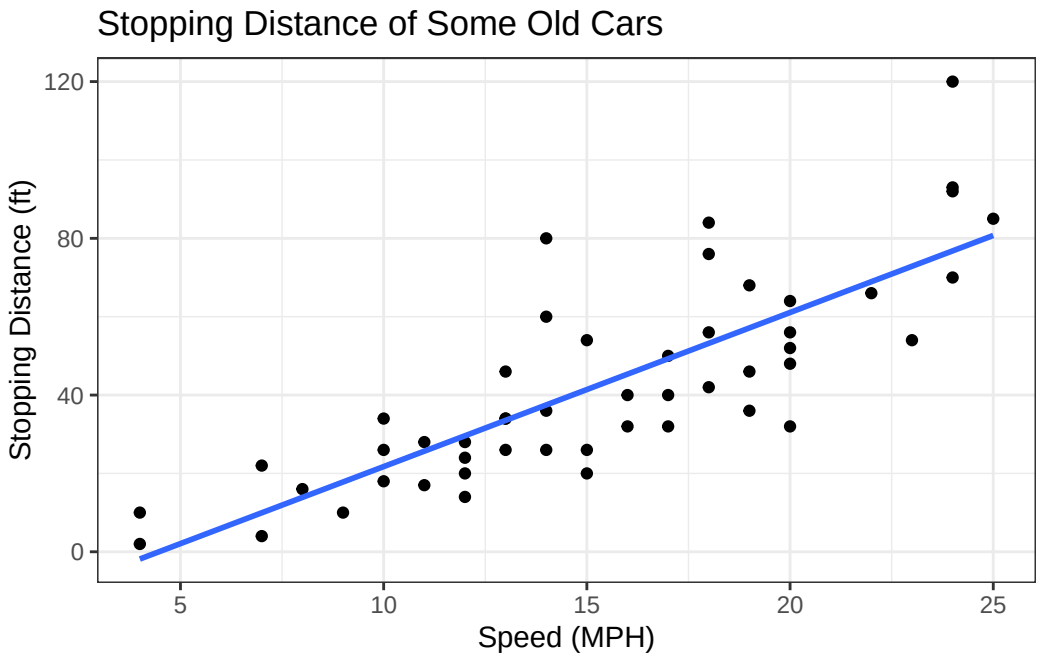


Let's add some features (more layers)

Let's add a trend line that shows the relationship between `speed` and `distance`

```
ggplot(cars) +
  aes(x = speed, y = dist) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_bw() +
  labs(x = "Speed (MPH)",
       y = "Stopping Distance (ft)",
  title= "Stopping Distance of Some Old Cars")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Gene Expression Plot

First we will read some expression data for the effects of a GLP-1 drug

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"

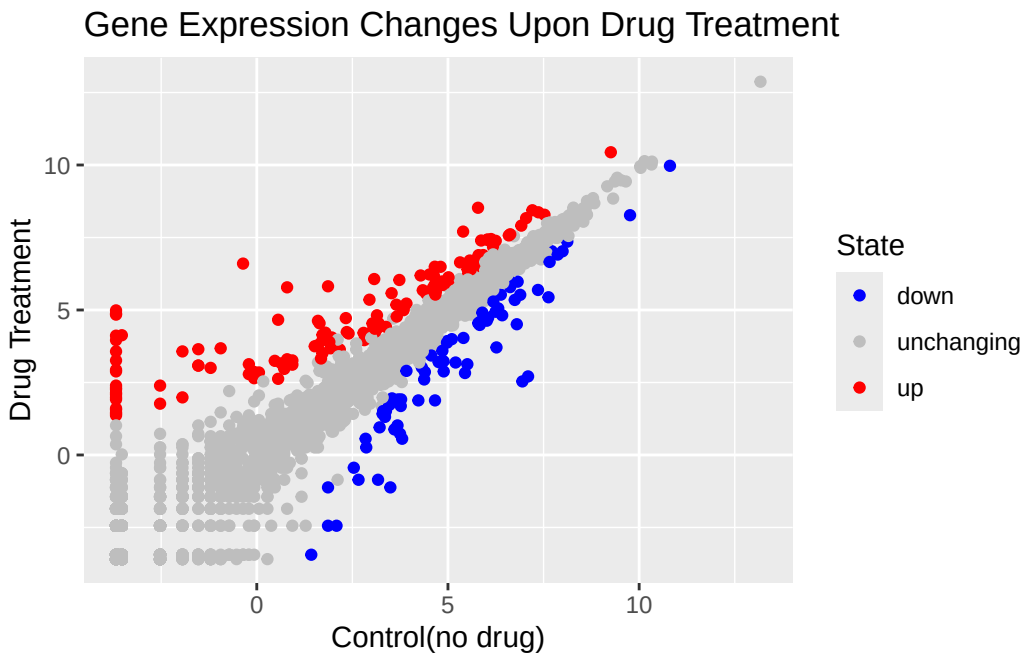
genes <- read.delim(url)

head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```
ggplot(genes) +
  aes(x = Condition1, y = Condition2, col = State) +
  geom_point() +
```

```
  scale_colour_manual(values = c("blue", "gray", "red")) +
  labs(x = "Control(no drug)",
       y = "Drug Treatment",
       title = "Gene Expression Changes Upon Drug Treatment")
```



Q. Use the nrow() function to find out how many genes are in this dataset.What is your answer?

```
nrow(genes)
```

```
[1] 5196
```

Q. Use the colnames() function and the ncol() function on the genes data frame to find out what the column names are (or will need these later) and how many columns there are. How many columns did you find?

```
colnames(genes)
```

```
[1] "Gene"       "Condition1" "Condition2" "State"
```

```
ncol(genes)
```

```
[1] 4
```

Q. Use the table() function on the State column of this data.frame to find out how many "up regulated genes there are. What is your answer?

```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

Q. Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset?

```
n.total <- nrow(genes)
n.up <- sum(genes$State == "up")
round((n.up/n.total*100), 2)
```

```
[1] 2.44
```

Going Further

Here we read the famous "gapminder" dataset:

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

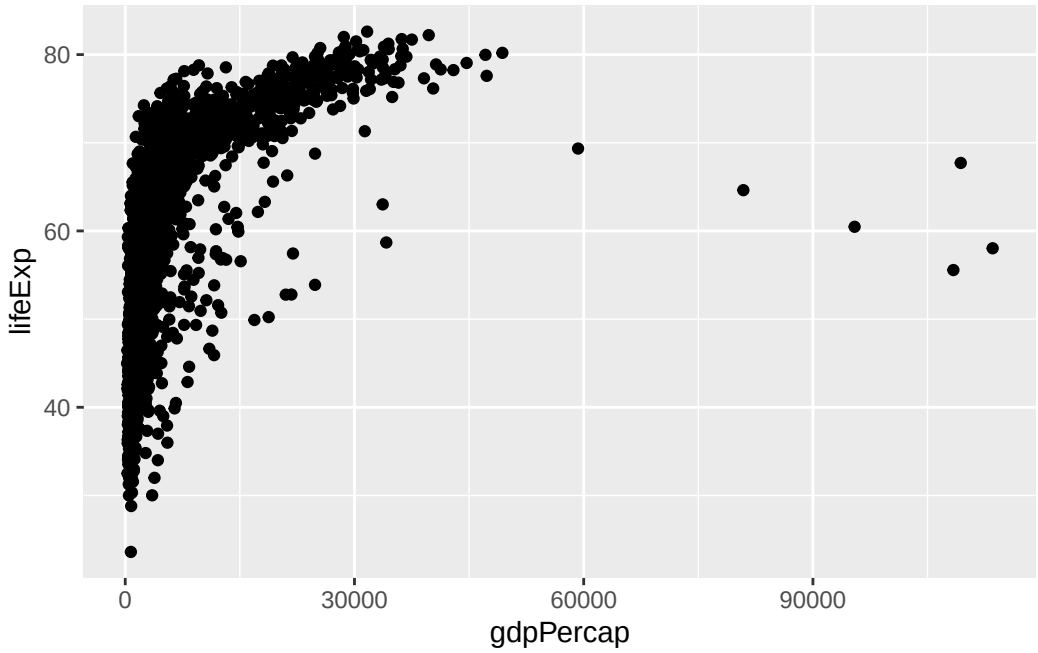
Q. How many different "country" values are in this dataset

```
length(table(gapminder$country))
```

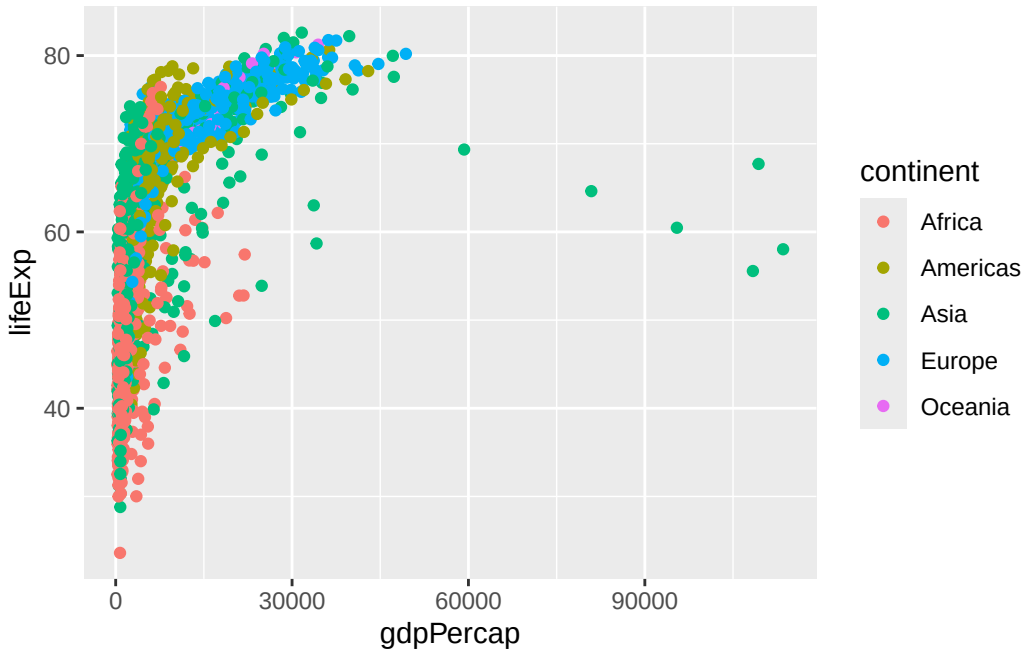
```
[1] 142
```

Lets make our first plot of “gdpPercap” vs “lifeExp”:

```
ggplot(gapminder) +
  aes(gdpPercap, lifeExp) +
  geom_point()
```

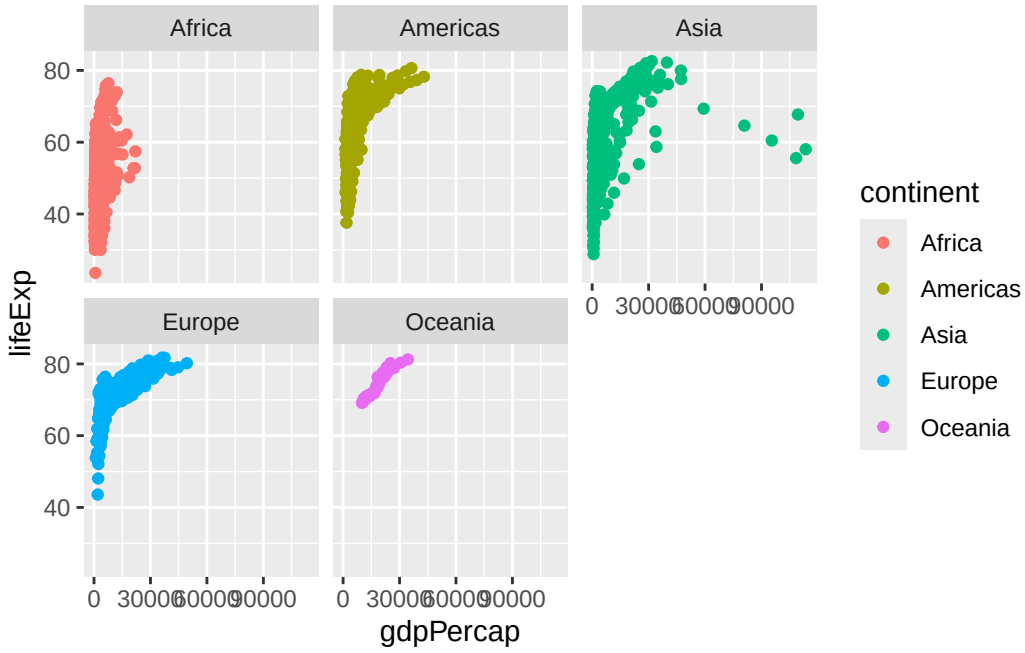



```
ggplot(gapminder) +
  aes(x = gdpPercap, y = lifeExp, col = continent) +
  geom_point()
```



One useful layer we can add to explore big details like this is called “faceting”

```
ggplot(gapminder) +
  aes(x = gdpPercap, y = lifeExp, col = continent) +
  geom_point() +
  facet_wrap(~continent)
```



Limit to 1977 and 2007

We can install and use the "dplyr" package to help us "filter" the full dataset to a smaller subset that only includes the years we want.

First we need to run `install.packages("dplyr")` and then load it with `library(dplyr)`.

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
filter(gapminder, year == 2007, country == "Ireland")
```

```
  country continent year lifeExp     pop gdpPercap
1 Ireland    Europe 2007  78.885 4109086     40676
```

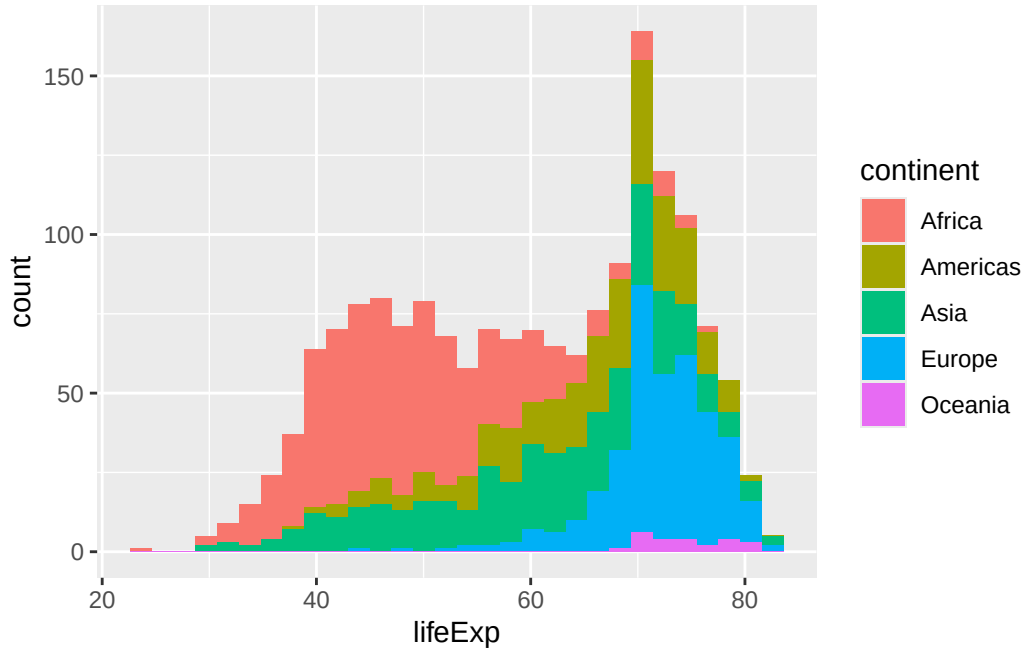
```
filter(gapminder, year == 1977, country == "Ireland")
```

```
  country continent year lifeExp     pop gdpPercap
1 Ireland    Europe 1977   72.03 3271900  11150.98
```

Q. Make a histogram of lifeExp fill color by continent

```
ggplot(gapminder) +
  aes(lifeExp, fill = continent) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Q. Make a boxplot of lifeExp fill color by continent

```
ggplot(gapminder) +
  aes(lifeExp, fill = continent) +
  geom_boxplot()
```

